



Customer Shopping Behavior Analysis

End-to-End Data Analytics Portfolio Project

PYTHON

SQL

POWER BI



Project Objective

Analyze Revenue

By gender, subscription status, and age group

Identify Loyalty

Which demographic drives strongest customer retention

Drive Decisions

Inform marketing, product, and retention strategy

Dataset Overview



The screenshot displays a web application interface for a retail dataset. On the left is a sidebar with navigation icons and labels in Japanese. The main area shows a table titled '楽天市場' (Rakuten Market) with 18 columns. The columns are: 来店 (Visit), SI/ (SI/), MI/ (MI/), AddOnCS, Customert, NISPS, RoG000, and Susimert. The table contains 10 rows of data, each representing a customer transaction. Each row includes a customer ID, a product name, a status indicator (green dot), a purchase amount, a payment amount, a discount amount, and a subscription status (green or orange dot).

来店	SI/	MI/	AddOnCS	Customert	NISPS	RoG000	Susimert
0020	0020	0020		\$983,2222	110,000	\$1,850	\$30 Y/H
0020	0020	0020		\$983,2222	110,000	\$1,850	\$30 Y/H
0020	0020	0020		\$983,2222	110,000	\$1,850	\$30 Y/H
0020	0020	0020		\$983,2222	110,000	\$1,850	\$30 Y/H
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0020	0020	0020		\$983,2222	110,000	\$1,850	\$30 Y/H

3,900 Customers · 18 Columns

Open-source retail dataset combining transactional and demographic data.

Demographics

Age, gender, location

Transactions

Purchase amount, payment, discount

Engagement

Subscription, review rating, frequency

Tools & Technology Stack



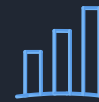
Python (pandas)

Data cleaning & transformation in Jupyter Notebook



SQL

10 queries analyzing customer behavior & revenue



Power BI

Interactive dashboard visualizing key findings

Project Workflow



Structured pipeline from raw CSV to interactive dashboard.

SQL Analysis – Key Queries

Revenue by Gender

Male **\$157,890** vs. Female **\$75,191**

Subscription Impact

Non-subscribers generate **\$170,436** vs. subscribers **\$62,645**; avg spend nearly identical

Customer Segments

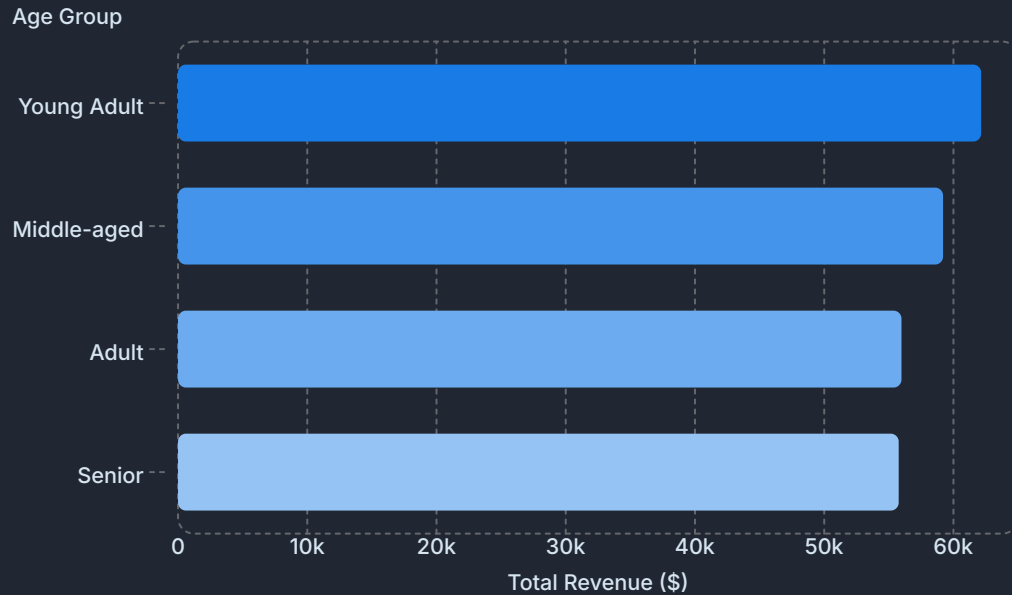
Loyal **3,116** · Returning **701** · New **83**

Discount-Reliant Products

Hats (50%), Sneakers (49%), Coats (49%), Sweaters (48%), Pants (47%)



Revenue by Age Group



Young Adults Lead

Young adults generate the highest revenue and represent the largest customer segment — a key demographic to prioritize.

Customer Behaviour Dashboard



Power BI Dashboard

3.9K
Customers

\$59.76
Avg Purchase

3.75
Avg Rating

27%
Subscribers



Key Findings

- **Male Revenue Dominance**
Driven by base composition, not individual spend
- **Weak Subscription Value**
Subscribers & non-subscribers spend similarly
- **Strong Baseline Loyalty**
Most customers are repeat buyers
- **Discount Dependency**
5 products rarely sell at full price

Business Recommendations

1

Women's Marketing

Invest in women's clothing & targeted campaigns

2

Revise Subscription

Add meaningful perks to boost sign-ups & retention

3

Stock Discount Items

Increase inventory of discount-reliant products during peak sale periods

4

Trend-Aligned Inventory

Keep offerings current for young adult demographic